

1.

Average: 590.624

Generation: 0

Average: 5610.64

Generation: 1

Average: 11597.72

Generation: 2

Average: 14859.104

Generation: 3

2.

I got a strategy that achieved over 50,000 (77192.0) on the average of 5 games in generation 2. It took a little less than 2 hours to fully run generations 0 to 2, each with a population size of 500.

3.

The best result, found in generation 3, gave a score of 131568.0 with the coefficients:

[0.19345767575542427, -0.9898862783635931, 0.9918290228393902, 0.7636453932151732, -0.35341614081732686, 0.042108393898946384, -0.060141583065019155].

The corresponding values of these coefficients are, in order, score gained, number of holes, highest point (larger heights are given by smaller numbers), number of interior blocks, lowest point (same note as highest point), number of total blocks, and the 2nd highest height difference between two columns.

Considering the scale of these possible values, the coefficients show that the most valued aspects were score gain, hole count, and the highest point. The number of interior blocks and deepest points were fairly valued, however total block count and 2nd highest height difference were valued very little.

Oddly, other strong strategies (such as the aforementioned 77192.0 strategy) placed greater emphasis on interior blocks, total blocks, and 2nd highest height difference while placing less value on the highest and lowest point.

I expected score gain, hole count, and highest point to be the most important factors. I thought that the 2nd highest height difference would have a greater impact, as performance increased heavily with its addition, but the best strategy places little value onto it. I expected the other three factors to be approximately their current positions in the best strategy.

4.

I liked this assignment because it is based around an interesting game - Tetris - and is a unique problem for using genetic algorithms. I thought that, while arduous at times, modeling games of Tetris and developing the genetic algorithm itself was interesting. I think that you should keep this assignment.